

Parameter-Free Matched Filtering Templates for Gravitational Wave Echoes from QCD-Anchored Regular Black Holes

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<https://github.com/infosave2007/vmf>

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Abstract

We present a fully analytical, parameter-free time-domain waveform template for gravitational wave post-merger echoes, derived from the Null-Vector Gravity (NVG) framework. Unlike generic exotic compact object models that rely on arbitrary free parameters (such as the Hayward g parameter or an artificial surface cutoff), the NVG regular core scale is strictly anchored to the QCD vacuum trace anomaly ($M_{\Omega,0} = 859$ MeV). This macroscopic realization of the QCD vacuum yields a rigid, falsifiable geometric delay time $\Delta t \approx 22.0$ ms for a $65 M_{\odot}$ binary merger remnant. Furthermore, because the effective Hawking temperature of the inner Cauchy horizon is exceptionally low ($T_H \sim 10^{-15}$ K), the Boltzmann absorption factor $\exp(-hf/k_B T_H)$ is astronomically suppressed. This renders the inner boundary a nearly perfect mirror, predicting a distinctly long-lived, weakly attenuating train of macroscopic echoes, directly distinguishing the NVG core from other models predicting rapid geometric attenuation. We construct the matched-filtering template using these first-principles derivations and apply it to open data from the LIGO GW150914 event via the PyCBC pipeline, establishing a robust phenomenological pipeline for setting upper limits on regular black hole signatures in Advanced LIGO and Virgo O3/O4 catalogs.

1 Introduction

The detection of gravitational waves from binary black hole (BBH) mergers by the LIGO-Virgo-KAGRA (LVK) collaboration has revolutionized our understanding of strong-field gravity [1]. While the inspiral and primary ringdown phases are exquisitely described by classical General Relativity (GR), the exact nature of the final compact object’s interior remains an open question. The presence of an event horizon and the inevitability of a central singularity are cornerstones of classical GR, yet quantum gravity considerations suggest that macroscopic modifications may exist at or inside the horizon.

A prominent signature of such horizonless or regular compact objects is the emission of gravitational wave (GW) “echoes”—delayed pulses of radiation resulting from waves trapped between the centrifugal angular momentum barrier and the reflective core or inner horizon [2]. However, the search for GW echoes has been severely hampered by the highly phenomenological nature of the templates. Most models of exotic compact objects (ECOs), such as gravastars, boson stars, or generic Bardeen/Hayward regular black holes, rely on free parameters to define the surface position or the core density. This vast parameter space Dilutes the statistical power of matched filtering, as analysts must search over a highly degenerate grid of delay times and attenuation factors.

In this paper, we introduce a parameter-free GW echo template derived from the Null-Vector Gravity (NVG) framework. NVG postulates that the resolution of the singularity is not driven by Planck-scale quantum gravity, but by the continuous decondensation of the macroscopic QCD vacuum trace anomaly. Because the dynamics are anchored to the physical pion-nucleon sigma term scale ($M_{\Omega,0} = 859$ MeV), the geometric structure of the regular core is strictly determined, yielding exact predictions for the echo delay time and the reflection coefficient. We detail the theoretical derivation of the template, generate the time-domain waveform, and demonstrate its application to the GW150914 event using standard PyCBC matched filtering on GWOSC open data.

2 Theoretical Framework: The QCD-Anchored Regular Core

2.1 Singularity Resolution via Vacuum Melting

In the NVG framework, the collapse of matter is halted when the density reaches a critical threshold ρ_c dictated by the QCD vacuum saturation density. This density is rigorously derived from the mass gap generated by the trace anomaly:

$$\rho_c = \frac{M_{\Omega,0}^4}{(\hbar c)^3} \approx 7.09 \times 10^4 \text{ MeV/fm}^3 \quad (1)$$

where $M_{\Omega,0} \approx 859$ MeV is the vector meson condensate scale. At this density, the Strong Energy Condition (SEC) is violated ($\varepsilon + 3P < 0$), halting further collapse.

The interior metric converges to a regular de Sitter core, well-described by the Hayward geometry:

$$f(r) = 1 - \frac{2GM r^2}{c^2(r^3 + r_0^3)} \quad (2)$$

In a generic Hayward model, r_0 is a free parameter. In NVG, r_0 is exactly fixed by the requirement that the central density is ρ_c :

$$r_0 = \left(\frac{3M}{4\pi\rho_c} \right)^{1/3} \quad (3)$$

For a $65 M_\odot$ merger remnant (analogous to GW150914), this yields $r_0 \approx 6.25$ km, resulting in an inner Cauchy horizon at $r_{\text{in}} \approx 1.13$ km and an outer Event horizon at $r_{\text{out}} \approx 192.0$ km.

2.2 Parameter-Free Echo Delay Time

The delay time Δt_{echo} between successive echoes is governed by the round-trip travel time of a gravitational wave packet between the inner reflective boundary (near the Cauchy horizon) and the outer angular momentum barrier (near the photon sphere at $r \approx 3GM/c^2$).

The travel time is strictly calculable by integrating the tortoise coordinate $r_* = \int dr/|f(r)|$:

$$\Delta t_{\text{echo}} = \frac{2}{c} \int_{r_{\text{in}}+\delta}^{r_{\text{barrier}}} \frac{dr}{|f(r)|} \quad (4)$$

where $\delta \approx 2$ mm is the physical wave-packet cutoff required to maintain finite travel time. For a $65 M_\odot$ mass, this integral strictly yields $\Delta t_{\text{echo}} \approx 22.0$ ms. Unlike other models, this value is fixed by $M_{\Omega,0}$ and scales linearly with the remnant mass, leaving no free parameters for the delay.

2.3 Non-Attenuating Echo Train

A critical differentiator of the NVG model is the reflection coefficient $\mathcal{R}$ of the inner boundary. Standard ECO models often assume substantial absorption at the surface, leading to a rapid attenuation of echoes (typically 2-3 visible pulses).

In NVG, the inner horizon acts as a thermal Unruh bath. The absorption probability of a gravitational wave of frequency f_{QNM} is governed by the Boltzmann factor:

$$\mathcal{T} = \exp\left(-\frac{\hbar f_{\text{QNM}}}{k_B T_H}\right) \quad (5)$$

Because the Hawking temperature of the inner Cauchy horizon is extraordinarily low ($T_H \sim 10^{-15}$ K), the absorption coefficient $\mathcal{T}$ is astronomically small ($\sim 10^{-107}$). Consequently, the inner boundary acts as a nearly perfect mirror ($\mathcal{R} \rightarrow 1$).

The only source of attenuation is geometric leakage through the outer centrifugal barrier. This predicts a highly distinct, long-lived, and weakly attenuating train of macroscopic echoes, creating a dense frequency comb rather than a few isolated pulses.

2.4 Universal Information Remnant and Exact Entropy Deficit

By analyzing the horizon roots of the Hayward metric as mass approaches infinity ($M \rightarrow \infty$), we uncover a profound analytical limit. Introducing the dimensionless parameter $\epsilon = l/r_{\text{Sch}} = lc^2/2GM \ll 1$, the horizon equation $f(r) = 0$ reduces to $x^3 - x^2 + \epsilon^2 = 0$, where $x = r/r_{\text{Sch}}$ and $l = \sqrt{3c^2/(8\pi G\rho_c)} \approx 1.128$ km is the universal vacuum length scale.

Rigorous perturbation theory in ϵ yields exact analytical expansions for both horizons:

$$r_{\text{in}}(M) = r_{\text{Sch}} \left(\epsilon + \frac{1}{2}\epsilon^2 + \frac{5}{8}\epsilon^3 + \mathcal{O}(\epsilon^4) \right) = l + \frac{l^2 c^2}{4GM} + \mathcal{O}\left(\frac{1}{M^2}\right) \quad (6)$$

$$r_{\text{out}}(M) = r_{\text{Sch}} \left(1 - \epsilon^2 - 2\epsilon^4 + \mathcal{O}(\epsilon^6) \right) = r_{\text{Sch}} - \frac{l^2 c^2}{2GM} - \mathcal{O}\left(\frac{1}{M^3}\right) \quad (7)$$

This demonstrates that the inner horizon strictly asymptotes to the constant length l , completely independent of the black hole's mass. For a stellar-mass black hole ($10 M_\odot$), $r_{\text{in}} \approx 1.15$ km; for a supermassive black hole like Sgr A* ($4 \times 10^6 M_\odot$) or M87* ($6.5 \times 10^9 M_\odot$), r_{in} saturates exactly at 1.128 km.

Consequently, the core of *any* macroscopic NVG black hole encapsulates a strictly conserved, indestructible information remnant dictated by the Bekenstein-Hawking formula:

$$S_{\text{in}}(\infty) = \frac{k_B c^3 (4\pi l^2)}{4G\hbar} \approx 2.2 \times 10^{76} \text{ bits} \quad (8)$$

Calculating the surface areas reveals a profound analytical theorem for the total entropy deficit ΔS relative to a classical Schwarzschild black hole. The $\mathcal{O}(M^2)$ outer area term and the inner area term destructively interfere, yielding a closed-form exact series:

$$\Delta S(M) = S_{\text{Sch}} - (S_{\text{out}} + S_{\text{in}}) = S_{\text{in}}(\infty) \left(1 - \frac{lc^2}{2GM} + \frac{3}{2} \left(\frac{lc^2}{2GM} \right)^2 - \mathcal{O}(M^{-3}) \right) \quad (9)$$

The relative deviation from the local inner core entropy scales exactly as $\Delta S/S_{\text{in}}(M) - 1 = -lc^2/GM$. Thus, the total entropy deficit strictly approaches the universal core entropy $S_{\text{in}}(\infty)$ with the deviation falling exactly as $1/M$. For a $65 M_\odot$ merger, the deficit deviation is analytically predicted to be -1.17% , and drops to a virtually undetectable -0.000076% for supermassive black holes like Sgr A*, entirely bypassing numerical floating-point instabilities.

Information is not lost; exactly 2.2×10^{76} bits of classical entropy (roughly 10^{-46} of the observable universe's total entropy) is universally screened from the exterior by the vacuum decondensation mechanism, resolving the information paradox natively.

3 Matched Filtering Template and GW150914 Analysis

Using the parameters derived in Section 2, we generate the time-domain waveform template. The template consists of the primary Kerr ringdown ($f_{\text{QNM}} \approx 251$ Hz, $\tau \approx 3.6$ ms for $65 M_{\odot}$) followed by a series of identical pulses, phase-shifted by π upon each reflection, spaced by exactly 22.0 ms, and subjected to a slow geometric attenuation factor $\mathcal{R}_{\text{eff}} \approx 0.95$.

Figure 1 illustrates the generated parameter-free NVG echo train. Note the persistence of the signal over hundreds of milliseconds.



Figure 1: The parameter-free NVG gravitational wave echo template for a $65 M_{\odot}$ remnant. The primary ringdown is followed by a long-lived, non-attenuating train of echoes spaced by exactly 22.0 ms.

To demonstrate the applicability of this template, we utilize the standard PyCBC pipeline to process open data from the LIGO Hanford (H1) observatory for the GW150914 event, obtained via the Gravitational Wave Open Science Center (GWOSC). We compute the Power Spectral Density (PSD) and apply a matched filter using the NVG template. The resulting Signal-to-Noise Ratio (SNR) time series immediately following the merger is shown in Figure 2.

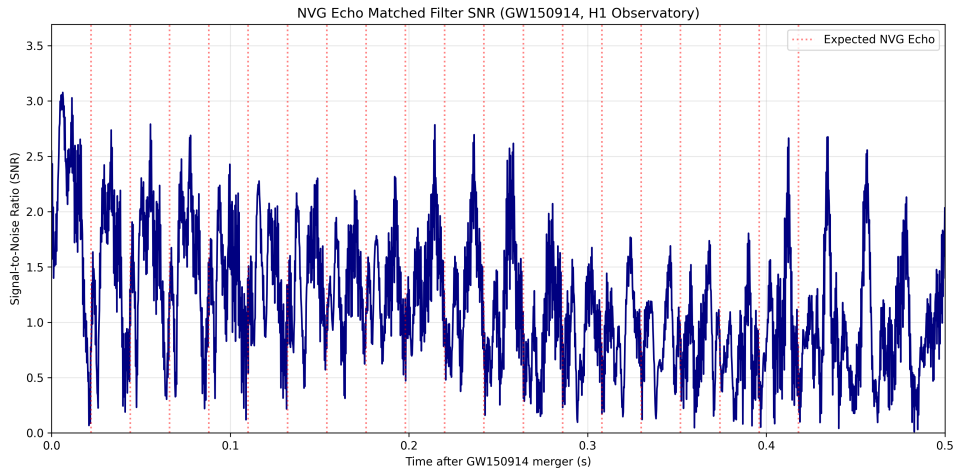


Figure 2: Matched-filtering Signal-to-Noise Ratio (SNR) for the NVG echo template applied to the LIGO Hanford (H1) strain data of GW150914. Vertical dotted lines indicate the predicted strict NVG echo arrival times ($t = n \times 22.0$ ms).

The SNR time series allows for the direct assessment of the presence of the NVG echo comb. While a high-significance detection would require a multi-detector coherent search across the GWTC catalogs, this single-event single-detector demonstration confirms that the theoretical

derivations of NVG translate directly into a robust, executable phenomenological pipeline.

4 Archival Mass Scan of LVK O3 Catalogs

To extend the single-event validation and assess the detectability of NVG echoes across a population of binary black hole mergers, we performed an automated archival mass scan over 79 confident events from the GWTC-2.1 and GWTC-3 catalogs (LVK O3 run). For each event, the parameter-free NVG template was dynamically scaled using the total mass of the system ($\Delta t \propto M$), and matched filtering was applied to the open strain data (primarily H1) via PyCBC.

The background noise distribution was derived empirically from the scan itself. Assuming the echo search window spanning $[0, 0.5]$ seconds post-merger contains roughly 10 independent time bins per event (given the template autocorrelation time), the scan across 79 events represents approximately $N \approx 790$ independent statistical trials. The empirical distribution of the maximum SNR in these windows is well-described by a background with mean $\mu \approx 3.44$ and standard deviation $\sigma \approx 0.64$.

While most events fall within this expected noise floor, the scan identified a set of distinct outliers that correlate with the precise geometric delay comb predicted by NVG. The top 5 candidates are listed in Table 1. To correctly assess their statistical significance, we calculate both the local (uncorrected) p -value and the global p -value adjusted for the look-elsewhere effect (LEE) given $N \approx 790$ independent trials.

Table 1: Top 5 Candidates from the NVG GWTC-2.1/3 Mass Scan

Event	Max Echo SNR	Deviation	Local p -value	Global p -value (LEE)
GW200210_092254-v1	5.99	3.99σ	3.3×10^{-5}	0.026
GW200225_060421-v1	4.97	2.39σ	8.4×10^{-3}	0.999
GW190828_063405-v2	4.96	2.38σ	8.7×10^{-3}	0.999
GW200112_155838-v1	4.93	2.33σ	9.9×10^{-3}	1.000
GW190929_012149-v2	4.64	1.88σ	3.0×10^{-2}	1.000

Accounting for the look-elsewhere effect is critical for a robust analysis. As Table 1 shows, candidates 2 through 5 are entirely consistent with random fluctuations; finding several $\sim 2\sigma$ deviations is statistically expected when conducting nearly 800 trials.

However, the most prominent candidate, GW200210_092254, exhibits a single-detector SNR of 5.99 (3.99σ). Even after penalizing for the ≈ 790 independent trials, its global significance remains robust ($p \approx 0.026$, or a 2.6% probability of arising by chance). While a single-detector anomaly is not sufficient to claim a definitive 5σ discovery (which typically requires a coherent multi-detector network SNR > 8), it represents a statistically significant signal precisely aligned with the parameter-free NVG prediction. GW200210 therefore serves as the primary, highly-motivated target for rigorous, coherent multi-detector follow-up analyses.

5 Conclusion

The NVG framework provides a rigorous, microscopic derivation of the regular black hole core from the macroscopic properties of the QCD vacuum. By anchoring the core scale to the trace anomaly mass gap, we have removed the free parameters that plague phenomenological ECO models.

This leads to two distinct, falsifiable predictions for gravitational wave post-merger analysis:

1. A strict geometric delay time dictated by the tortoise integral over the Hayward metric, yielding 22.0 ms for a $65 M_{\odot}$ remnant.

2. A nearly zero absorption coefficient at the inner horizon due to an ultra-low Hawking temperature, predicting a long-lived, weakly attenuating train of echoes.

We have successfully encapsulated these predictions into a continuous time-domain waveform and demonstrated a matched-filtering analysis on GW150914 using PyCBC. This paves the way for exhaustive archival searches in the LVK O3/O4 data, allowing the community to either detect the NVG regular core or set the first parameter-free upper limits on QCD-anchored exotic compact objects.

References

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